Moscow Institute of Physics and Technology

Game theory, Spring 2026
Problem set 1 (static games), due date: April 7

1. **(Two games 2×2)** Find all pure and mixed Nash equilibria in the following games:

   • **(Meeting in mountains)** Two hiking groups (the Red and the Blue) choose where to cross the mountain ridge: through the Northern pass or through the Southern pass. They start from the different sides of the ridge and do not have any connection to each other. If they both choose the Northern pass, then they meet at the pass with probability 90%. If they both choose the Southern pass, then they meet with probability 70%. If they choose different passes, they do not meet at all, of course. The Red get 10 units if they meet, and the Blue get 20 units in this case. For the Red, the cost of going through the Northern pass is 5 and the cost of going through the Southern pass is 7. For the Blue, the costs are 10 for the Northern pass and 8 for the Southern pass.

   Draw the matrices and solve the game. (The players are interested only in the expected profit).

   • **(Penalty kicks)** A Forward and a Goalkeeper play a penalty kick. The Forward can shoot to the left corner or to the right corner. If he shoots on the right, then he is always on target. But if he shoots on the left, then he is on target only with probability $\frac{5}{6}$. The Goalkeeper can jump to the left corner or to the right corner. If he jumps to the left and the ball is also on the left, then he always catches the ball. But if he jumps to the right, then he catches the ball only with probability $\frac{2}{3}$. If the ball and the Goalkeeper are in different corners, then the Goalkeeper never catches the ball.

   The Forward gets 6 points if he hits the target, -6 points if he misses the target and 0 points if the Goalkeeper catches the ball. The Goalkeeper get 6 points if he catches the ball, -6 points if the Forward hits the target and 0 points if the Forward misses.

   Draw the matrices and solve the game. (The players are interested only in expected outcome).

2. **(Parametric game 2×3)** Consider the following game with parameter a:

	L	M	R
U	4, 1	3, 6	a, 7
D	5, 6	2, 5	5, 3

   Find all pure and mixed Nash equilibria in cases:

   (a) $a = 3$;

   (b) $a = 6$;

   (c) $a = 5$.

3. **(Abstract game)** Consider the following game:

	t_1	t_2	t_3	t_4	t_5
s_1	2, 0	5, 3	2, 4	6, 1	3, 2
s_2	3, 1	0, 6	0, 5	3, 2	0, 3
s_3	4, 6	1, 3	2, 1	5, 4	1, 3
s_4	2, 7	4, 2	1, 4	5, 8	4, 5

   (a) By eliminating strictly dominated strategies reduce the game to the form $2 \times N$. Specify which strategy is dominated by which one. Note that a strategy may be dominated not only by a particular strategy but also by a mixture of strategies. Any further eliminations must be impossible in the remaining game.

   (b) Find all pure and mixed equilibria in the remaining $2 \times N$ game.

4. **(Normandy landings)** In 1944, the Allies invaded Normandy, that influenced the outcome of World War II. The Allies had 6 possible attacking configurations (1–6) and the Nazis had 6 possible defensive strategies (A–F). All possible battles were simulated and the following outcomes for the Allies were estimated (the outcomes for the Nazis were the negations of these values).

	A	B	C	D	E	F
1	13	29	8	12	16	23
2	18	22	21	22	29	31
3	18	22	31	31	27	37
4	11	22	12	21	21	26
5	18	16	19	14	19	28
6	23	22	19	23	30	34

   (a) Eliminate all dominated strategies.

   (b) Try to solve the remaining game.

   (c) In reality, the Germans chose defense B and the Allies chose attack 1. Comment on these choices.

5. **(Condorcet voting)** George, Ferdinand and Irene choose where to fly on vacations. There are three variants: Germany, France and Italy. George prefers Germany, his second choice is France, and the least preferable option is Italy, Ferdinand prefers France, then Italy, then Germany. Finally, Irene prefers Italy, then Germany, then France. They adopt the following procedure: each one names an option, if any option is named by two or three people then it is chosen, and if all name different options then the option named by Irene is chosen.

   (a) Formulate the situation as a one-shot game;

   (b) Find an equilibrium by eliminating weakly dominated strategies;

   (c) Find all Nash equilibria.

6. **(Lazy voters)** There are n voters who choose among two alternatives: C and D. There are k voters who support C and all others support D. Every voter chooses whether to participate in the referendum. Participation costs 1 unit for every voter. Of course, a participating voter casts a vote in favor of his preferred option. If finally one alternative gets more vote than the other, then all supporters of the winning option (participating and non-participating) get 4 units. If there is a tie, then all voters get 2 units.

   Find all pure Nash equilibria in the following cases:

   (a) $k < \frac{n}{2}$;

   (b) $k = \frac{n}{2}$.

7. **(Battleship in a canal)** Two boys are playing Battleship on a 1×4 field with a 1×2 ship. The first boy places the ship and the second one attacks it.

   (a) Suppose that the attacker wins if he strikes the ship with the first missile. Formulate the situation as a one-shot game and find all Nash equilibria.

   (b) Suppose that the attacker wins if he kills the ship with the first two missiles. Formulate the situation as a one-shot game and find all Nash equilibria.

8. **(A lottery with two cars)** A TV show organizes a lottery with two cars as prizes. Any watcher may send an SMS for participating. If there are only one or two participants then they get a car. Otherwise nobody wins. Find a symmetric mixed Nash equilibrium. What is the probability that any car is won? (Think that n is very large)

9. **(A public hall is never swept)** A man gets a heart attack at a metro station. Every observer chooses whether to call emergency or just to pass by. They make decisions simultaneously and independently. If at least one observer calls then the man will be saved, otherwise he will die. Calling emergency costs a small positive ε. If the man dies then all observers get disutility -1. Find a symmetric mixed Nash equilibrium. How does it change when n rises?